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Ultimate Amiga OS — DOS & Scripting Manual

Version 0.1.0-dev | Ultimate Amiga OS Shell Reference

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Introduction

The UAOS shell is a hybrid AmigaDOS-style command interpreter that runs natively on the x86_64 kernel. It combines traditional AmigaDOS conventions (assigns, C:, S:, ENV: paths) with modern bare-metal capabilities including TCP/IP networking, block-device management, and script flow control.

Commands are case-insensitive. Both DIR and dir work identically.

Command Types

Type	Description	Examples
Shell Built-Ins	Executed directly by the shell; manage shell state	cd, alias, set
C: Binaries	Native x86_64 commands registered in native_cmd.c	dir, copy, ping
M68k Binaries	Legacy Amiga Hunk binaries run via emulation	any file found in PATH

Getting Started

The Prompt

When you open a Shell window you see:

```
UAOS>
```

Type commands and press **Enter** to execute.

Command History

- **Up Arrow** — recall previous command
- **Down Arrow** — recall next command
- **Tab** — auto-complete filenames and command names

Stopping a Command

There is no dedicated break key. Commands that loop (e.g. **ping**) run a fixed number of iterations and yield automatically to keep the desktop responsive.

Shell Built-In Commands

These commands are handled directly by the shell and are not separate binaries on **C:**.

help

Show the complete list of built-ins and C: binaries.

```
UAOS> help
```

cd [path]

Change working directory, or print the current directory if no argument is given.

```
UAOS> cd Workbench:C
```

```
UAOS> cd RAM:T
```

```
UAOS> cd ..          ; parent directory
```

```
UAOS> cd              ; prints current directory
```

alias [name cmd]

Create, update, or list command aliases.

```
UAOS> alias ll "dir"
```

```
UAOS> alias          ; list all aliases
```

```
UAOS> alias h "help"
```

unalias <name>

Remove an alias.

```
UAOS> unalias ll
```

set [name value]

Set or list **local** environment variables (per-shell, not persisted).

```
UAOS> set greeting "Hello World"
```

```
UAOS> set                ; list all local variables
```

unset <name>

Remove a local environment variable.

```
UAOS> unset greeting
```

path [dirs...]

Show or set the command search path. The path is a space-separated list of directories searched when a command is not a built-in or native binary.

```
UAOS> path                ; show current path
```

```
UAOS> path C: S:          ; set path to C: and S:
```

setenv <name> <value>

Set a **global** environment variable. Writes to both the local shell store and ENV:<name> so it persists across sessions.

```
UAOS> setenv Language "english"
```

unsetenv <name>

Remove a global environment variable from both the local store and ENV:.

```
UAOS> unsetenv Language
```

rename <from> <to>

Not yet implemented. Use copy followed by delete as a workaround.

C: Binaries (Native Commands)

All commands below are native x86_64 binaries located conceptually on C:. They are dispatched directly by the kernel without M68k emulation.

System Information

version Display OS version and hardware summary.

```
UAOS> version
```

```
Ultimate Amiga OS  v0.1.0-dev
```

```
Kernel: x86_64 ELF64, Multiboot2, long mode
```

```
Display: 1024x768 32bpp linear framebuffer
```

```
Input: PS/2 keyboard + mouse, IRQ1/IRQ12
```

showconfig Show hardware configuration in Amiga-style format (processor, RAM, boards).

```
UAOS> showconfig
```

mem Display memory layout.

```
UAOS> mem
RAM: 512 MB (QEMU)
Kernel load: 0x0000000000100000
Framebuffer: mapped (GOP physical address)
Stack: 16 KB (bootstrap), no heap allocator yet
```

libs List all loaded kernel ROM libraries with versions.

```
UAOS> libs
```

date Display current local date and time. When **ntpd** has run, shows timezone-aware local time. Otherwise falls back to CMOS RTC (UTC).

```
UAOS> date
Saturday 14-Jun-2026 23:12:57 AEST (Australia/Sydney)
```

info [device] Show mounted disks and volumes. With an argument, show details for a specific device.

```
UAOS> info
UAOS> info RAM:
UAOS> info DH0:
```

disks List detected block devices (whole disks only).

```
UAOS> disks
```

which <cmd> Locate a command: reports if it is a shell built-in, a **C:** binary, or found in the **PATH**.

```
UAOS> which dir
c:dir
UAOS> which cd
cd is a shell built-in command
```

Filesystem Commands

dir [path] List directory contents. If no path is given, lists the current directory.

```
UAOS> dir
UAOS> dir C:
UAOS> dir RAM:T
```

makedir <path> Create a new directory.

```
UAOS> makedir RAM:MyDir
```

delete <path> Delete a file or an **empty** directory.

```
UAOS> delete RAM:T/old.txt
```

type <file> Print the contents of a file to the shell.

```
UAOS> type S:Startup-Sequence
```

copy <src> <dst> Copy a file.

UAOS> copy S:Startup-Sequence RAM:T/backup.txt

rename <from> <to> Not yet implemented. Use copy then delete.

protect [+r|-r][+h|-h] <path> Set file protection attributes.

Flag	Meaning
+r	Set Read-Only
-r	Clear Read-Only
+h	Set Hidden
-h	Clear Hidden

UAOS> protect +r RAM:important.txt

UAOS> protect -h +r RAM:config.dat

attr <path> Display file or directory attributes.

UAOS> attr RAM:important.txt

Attributes: Read-Only

pwd Print the current working directory.

UAOS> pwd

Workbench:SYS

echo <text> Print text to the shell. Supports variable expansion with \$var.

UAOS> echo "Hello World"

UAOS> echo "User is \$User"

Disk Management

fdisk <device> Interactive partition table editor. Enter interactive mode with commands m (help), n (new), d (delete), p (print), w (write), q (quit).

UAOS> fdisk virtio0

UAOS> fdisk -l ; list available disks

format <device> [filesystem] Format a partition (not a whole disk). Supports fat32.

UAOS> format virtio01 fat32

UAOS> format Device=DH0: Name=Workbench FFS

Networking Commands

ifconfig [dhcp | <ip> <gateway>] Show or configure network settings. /24 netmask is assumed.

UAOS> ifconfig ; show config

UAOS> ifconfig 192.168.1.10 192.168.1.1

UAOS> ifconfig dhcp ; run DHCP discovery

ping <host> [count] Send ICMP echo requests. Default count is 4, max is 64.

```
UAOS> ping 8.8.8.8
```

```
UAOS> ping google.com 8
```

route Display the kernel routing table and ARP cache.

```
UAOS> route
```

nslookup <hostname> [server] Resolve a hostname to an IP address using the configured DNS server.

```
UAOS> nslookup google.com
```

```
UAOS> nslookup google.com 8.8.8.8
```

ntpd [server] One-shot NTP time synchronisation. Reads **S:ntp.conf** and **S:timezone.conf**.

```
UAOS> ntpd
```

```
UAOS> ntpd pool.ntp.org
```

Text Processing

grep [-i] <pattern> <file> Search file contents for a pattern. Use **-i** for case-insensitive matching.

```
UAOS> grep error S:Startup-Sequence
```

```
UAOS> grep -i amiga RAM:notes.txt
```

more <file> Paginate file output one screen at a time.

Key	Action
Space / Page-Down	Advance one page
Enter	Advance one line
q / Escape	Quit

```
UAOS> more S:Startup-Sequence
```

Desktop & Tools

clear Clear the shell window and history buffer.

```
UAOS> clear
```

reboot Warm reboot via the keyboard controller.

```
UAOS> reboot
```

pointer Open the Pointer Preferences window.

```
UAOS> pointer
```

calculator Open the Calculator window.

```
UAOS> calculator
```

clock Open the Clock window.

```
UAOS> clock
```

loadwb Launch the Workbench desktop.

```
UAOS> loadwb
```

Amiga Compatibility

run <program> [args] Execute an embedded Amiga M68k binary from the ROM registry.

```
UAOS> run Calculator
```

assign [name: target] Create or list AmigaDOS assigns. With no arguments, lists current assigns.

```
UAOS> assign                ; list assigns
```

```
UAOS> assign C: Workbench:C
```

```
UAOS> assign ENV: RAM:ENV
```

execute <script> Run a script file line by line with full flow-control support.

```
UAOS> execute S:Startup-Sequence
```

```
UAOS> execute RAM:T/myscript
```

Scripting & Flow Control

UAOS scripts are plain text files executed line-by-line via the **execute** command (or the **execute** shell built-in). Lines beginning with **;** or ***** are treated as comments.

Single-Line IF

```
IF <condition> THEN <command>
```

Examples:

```
IF EXISTS S:timezone.conf THEN echo "Config found"
```

```
IF $count EQ 5 THEN echo "Five"
```

```
IF NOT EXISTS RAM:lock THEN echo "No lock file"
```

Multi-Line IF / ELSE / ENDIF

```
IF <condition>
    <commands...>
```

```
ELSE
    <commands...>
```

```
ENDIF
```

Example:

```
IF EXISTS S:timezone.conf
    echo "Timezone configured"
    C:ntpd
```

```
ELSE
    echo "No timezone config"
ENDIF
```

FOR / ENDFOR

```
FOR <var> = <start> TO <end> [STEP <step>]
  <commands...>
ENDFOR
```

Examples:

```
FOR i = 1 TO 3
  echo "Loading module $i..."
ENDFOR
```

```
FOR count = 10 TO 0 STEP -2
  echo "Countdown: $count"
ENDFOR
```

Conditions

Condition	Meaning
EXISTS <file>	True if file or directory exists
<a> EQ 	Strings are equal
<a> NE 	Strings are not equal
NOT <condition>	Logical negation

Note: Conditions are evaluated left-to-right. Variable references with **\$var** are expanded before the condition is tested.

Environment Variables

Local Variables (set / unset)

Stored in the shell instance. Lost when the shell closes.

```
Set greeting "Hello"
echo "$greeting World"
```

Global Variables (setenv / unsetenv)

Stored in ENV: as files. Persist across sessions.

```
SetEnv Language "english"
UnSet Language
```

Pre-Defined Variables

Variable	Source
\$Workbench	Set during startup (then unset)
\$Kickstart	Set during startup (then unset)
Any ENV: file	Readable as \$name

Variable Expansion

Use `$name` or `${name}` (simple prefix expansion is supported). Variables are expanded: - In `echo` arguments - In IF condition strings - In general command lines before execution

```
Set user "Alice"
echo "Hello, $user"
IF EXISTS ENV:Language THEN echo "Language is set"
```

File Assigns

AmigaDOS assigns map logical volume names to physical paths. They are essential for Amiga compatibility.

Kernel-Default Assigns

Assign	Target
C:	Workbench:C
S:	Workbench:S
L:	Workbench:L
DEVS:	Workbench:DEVS
LIBS:	Workbench:LIBS
SYS:	Workbench:SYS

Common Startup Assigns

```
Assign ENV: RAM:ENV
Assign T: RAM:T
Assign Clips: RAM:Clipboards
Assign REXX: S:
Assign PRINTERS: DEVS:Printers
Assign KEYMAPS: DEVS:Keymaps
Assign LOCALE: SYS:Locale
Assign LIBS: SYS:Classes ADD
```

Creating Your Own

```
Assign MyFiles: RAM:MyStuff
```

I/O Redirection

The shell supports three redirection operators:

Operator	Description
<code>> file</code>	Overwrite output to file
<code>>> file</code>	Append output to file
<code>< file</code>	Read input from file (placeholder; not all commands support it)

```
dir > RAM:listing.txt
echo "Log entry" >> RAM:log.txt
version >NIL:           ; suppress output
```

Note: >NIL: and similar pseudo-assigns are treated as literal paths unless NIL: is explicitly mounted.

Example Scripts

Basic Startup Script

```
; Ultimate Amiga OS - Startup-Sequence
C:Version >NIL:

Assign >NIL: ENV: RAM:ENV
Assign >NIL: T: RAM:T

SetEnv Language "english"

; Welcome message
echo "Ultimate Amiga OS - Live Workbench Environment"
echo ""

; Demo: conditional execution
IF EXISTS S:timezone.conf
    echo "Timezone configuration present"
ENDIF

; Demo: numeric loop
FOR i = 1 TO 3
    echo "Loading system module $i..."
ENDFOR

; Synchronise system clock
C:ntpd

; Load Workbench desktop
echo ""
echo "Starting Workbench..."
C:LoadWB
```

Backup Script

```
; backup.script - copy important files to RAM:
IF NOT EXISTS RAM:Backup
    mkdir RAM:Backup
ENDIF

copy S:Startup-Sequence RAM:Backup/
copy S:ntp.conf RAM:Backup/
copy S:timezone.conf RAM:Backup/

echo "Backup complete."
```

Network Test Script

```
; nettest.script - verify network connectivity
IF EXISTS ENV:Server
    echo "Testing server $Server..."
```

```

ping $Server 4
ELSE
    echo "SetEnv Server <address> first"
ENDIF

```

Directory Scanner

```

; scan.script - list and count files
echo "Scanning C:..."
dir C: > RAM:dirlog.txt
grep dir RAM:dirlog.txt
echo "Scan saved to RAM:dirlog.txt"

```

Quick Reference Card

Built-Ins

Command	Syntax	Purpose
cd	cd [path]	Change directory
alias	alias [name cmd]	Manage aliases
unalias	unalias <name>	Remove alias
set	set [name value]	Local variables
unset	unset <name>	Remove local var
path	path [dirs...]	Search path
setenv	setenv <name> <value>	Global variable
unsetenv	unsetenv <name>	Remove global var
execute	execute <script>	Run script

Native Commands

Command	Syntax	Purpose
version	version	OS version
showconfig	showconfig	Hardware info
mem	mem	Memory layout
libs	libs	ROM libraries
date	date	Current date/time
info	info [dev]	Disk/volume info
disks	disks	Block devices
which	which <cmd>	Locate command
dir	dir [path]	List directory
makedir	makedir <path>	Create directory
delete	delete <path>	Delete file/dir
type	type <file>	Print file
copy	copy <src> <dst>	Copy file
protect	protect [+<->][rh] <path>	Set attributes
attr	attr <path>	Show attributes
pwd	pwd	Working directory
echo	echo <text>	Print text
fdisk	fdisk <dev> / fdisk -l	Partition editor
format	format <dev> [fs]	Format partition
ifconfig	ifconfig [dhcp ip gw]	Network config

Command	Syntax	Purpose
ping	ping <host> [count]	ICMP echo
route	route	Routing/ARP table
nslookup	nslookup <host> [srv]	DNS lookup
ntpd	ntpd [server]	NTP sync
grep	grep [-i] <pat> <file>	Search file
more	more <file>	Paginated view
clear	clear	Clear shell
reboot	reboot	Warm reboot
pointer	pointer	Pointer prefs
calculator	calculator	Calculator win
clock	clock	Clock win
loadwb	loadwb	Launch Workbench
run	run <prog>	M68k binary
assign	assign [name: tgt]	Create/list assigns
execute	execute <script>	Run script file

Script Keywords

Keyword	Syntax
IF	IF <cond> THEN <cmd> or block with ELSE / ENDIF
FOR	FOR <v> = <a> TO [STEP <s>] ... ENDFOR
Comments	; comment or * comment

End of Manual — Ultimate Amiga OS v0.1.0-dev